THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 70	Area $= \frac{1}{2} \times a \times b$.
2	**(c)** 25	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
3	**(d)** $\frac{7}{9}$	Each is $\frac{d}{9}$.
4	**(b)** 35	$x^3 + y^3 = (x+y)^3 - 3xy(x+y)$.
5	**(a)** 31	Add the two distances from the centre.
6	**(c)** 96	Use Brahmagupta's formula.
7	**(c)** $\frac{1}{9}$	List all pairs with the given product.
8	**(d)** 50	$d = 3$, $a = 5$; $t_{16} = a + 15d$.
9	**(c)** 21	Squares have an odd number of factors; count the integers whose squares lie in the range.
10	**(d)** 187	$34(78 - 111) = n_2(72 - 78)$.
11	**(c)** 10	Trace the sequence step by step.
12	**(d)** 15°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
13	**(b)** 153	Intercepts 18 and 17; area $= \frac{1}{2} \times 18 \times 17$.
14	**(d)** 36 cm	$\sqrt{l^2 + b^2 + h^2} = \sqrt{144 + 576 + 576}$.
15	**(d)** 5	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
16	**(b)** 4	Equate the two expressions.
17	**(d)** $\frac{4}{9}$	Add the numerators over 99.
18	**(c)** 98	$x^3 - y^3 = (x-y)^3 + 3xy(x-y)$.
19	**(c)** 17	Subtract the two distances from the centre.
20	**(a)** 42	Side of square $= 2r$.
21	**(c)** $\frac{2}{9}$	Count the favourable pairs out of 36.
22	**(b)** 37	$t_k = S_k - S_{k-1} = p(2k-1) + q$.
23	**(c)** 11	$n = 11$ gives 121, which is composite.
24	**(b)** 7 : 9	Amounts are in the ratio $(41-34) : (34-25) = 7 : 9$.
25	**(b)** 14	Pairs: $(1344, 872) \to (872, 472) \to (472, 400) \to (400, 72) \to (328, 72) \to (256, 72) \to (184, 72) \to (112, 72) \to (72, 40) \to (40, 32) \to (32, 8) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$.
26	**(c)** 70 cm	Sides 15 cm and 20 cm.
27	**(b)** 1101	Rate $= \dfrac{182}{7} = 26$ per km; fixed charge 113.
28	**(b)** 22176 cm^3	Volume of hemisphere + volume of cone.
29	**(b)** 41	Radius $= \frac{1}{2}\times$ distance between the endpoints.
30	**(b)** 25	Evaluate the inner value first.
31	**(b)** 3	Cancel every prime factor other than 2 and 5.
32	**(a)** 18	$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$.
33	**(a)** 14	Add the distances of the centres from the common chord.
34	**(b)** 770	Difference of the two areas.
35	**(c)** $\frac{7}{8}$	Use the complement rule where it is quicker.
36	**(a)** 816	AP with $a = 6$, $d = 6$.
37	**(c)** 3	$n = 3$ gives 203, which is composite.
38	**(a)** 32	$\dfrac{2 \times 20 \times 80}{20 + 80} = 32$ (total distance / total time).

39	**(c)** 8	Pairs: $(912, 720) \to (720, 192) \to (528, 192) \to (336, 192) \to (192, 144) \to (144, 48) \to (96, 48) \to (48, 48) \to (48, 0)$.
40	**(b)** 72 cm	Side $= 18$ cm.
41	**(a)** Both A and R are true and R is the correct explanation of A.	$x = -3 < 0$ and $y = 4 > 0$, which is exactly the sign pattern of Quadrant II.
42	**(a)** Both A and R are true and R is the correct explanation of A.	$-\dfrac{-12}{3} = 4$; R is the general rule that gives it.
43	**(a)** Both A and R are true and R is the correct explanation of A.	$0.375 = \dfrac{375}{1000} = \dfrac{3}{8}$; R is the general reason.
44	**(a)** Both A and R are true and R is the correct explanation of A.	$103^2 = (100 + 3)^2 = 10000 + 600 + 9 = 10609$, using R.
45	**(a)** Both A and R are true and R is the correct explanation of A.	Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm, using the perpendicular bisection in R; chord $= 8$ cm.
46	**(a)** Both A and R are true and R is the correct explanation of A.	$s = 21$, area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$, using R.
47	**(a)** Both A and R are true and R is the correct explanation of A.	$P = \dfrac{3}{6} = \dfrac{1}{2}$, obtained exactly as in R.
48	**(a)** Both A and R are true and R is the correct explanation of A.	$5 - 2 = 8 - 5 = 11 - 8 = 3$; a constant common difference is the definition of an AP.
49	**(a)** Both A and R are true and R is the correct explanation of A.	The converse is 'If a number is even, then it is divisible by 6'; 4 is a counterexample, which is exactly R.
50	**(a)** Both A and R are true and R is the correct explanation of A.	$\dfrac{20 \times 60 + 30 \times 70}{50} = \dfrac{3300}{50} = 66$, using R.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** 38	The middle number is the average, 36.
2	**(b)** 69	New salary is higher by $n \times 2 = 44$ thousand.
3	**(b)** 2075	The algorithm reverses the digits.
4	**(b)** 90 cm	Half-diagonals and the side form a right triangle: half of the other diagonal $= \sqrt{51^2 - 24^2} = 45$.
5	**(b)** 17	Equal slopes between consecutive points.
6	**(c)** 144	$\dfrac{54 \times 12 \times 6}{27}$.
7	**(c)** 4	Solve $(x - x_0)^2 + b^2 = c^2$ and take the larger root.
8	**(b)** 7	$kz - c = 0$.
9	**(a)** $\frac{14}{11}$	Write as $1 + 0.\overline{ab}$.
10	**(c)** 21	$2xy = (x + y)^2 - (x^2 + y^2)$.
11	**(a)** 108°	Linear pair at D.
12	**(c)** 120	Diagonals bisect at right angles: the other diagonal is $2\sqrt{s^2 - (d/2)^2}$.
13	**(d)** 21	Blue $=$ total $-$ red.
14	**(d)** 14	Take $m - d,\ m,\ m + d$; the sum gives the middle number.
15	**(a)** 1	$7^2 = 49$ and $49 = 3 \times 16 + 1$.
16	**(b)** 100	Sum of the extra numbers $= 88 \times 100 - 48 \times 75 = 5200$; divide by 52.
17	**(c)** 12218	LCM $= \dfrac{82 \times 298}{2}$.
18	**(c)** 1344 cm^2	Area $= \frac{1}{2} d_1 d_2$.
19	**(c)** -4	$8(1) + 2(4) = 16$ gives $k = 8$; slope $= -\frac{k}{2} = -4$.
20	**(c)** 45	Cylinder volume ÷ cone volume $= \frac{3h}{H}$.
21	**(a)** $(3, 4)$	$D = A + C - B$.
22	**(b)** $b = 0$	Reason about linear polynomials.
23	**(c)** 1	Reason about decimal expansions.
24	**(b)** 28	$2xy = (x^2 + y^2) - (x - y)^2$.
25	**(b)** 132°	Opposite angles of a cyclic quadrilateral.
26	**(a)** 8	The shortest altitude stands on the longest side.
27	**(d)** $\frac{7}{11}$	Count with repetition.
28	**(a)** 12100	Yearly values form a GP with ratio $1 + \frac{r}{100}$.
29	**(d)** 8	$10a + b - (a + b) = 9a$, so the quotient is the tens digit.
30	**(a)** 22%	$\dfrac{2000 \times 13 + 9000 \times 24}{11000} = 22\%$.
31	**(d)** 48	$\lfloor \sqrt{2381} \rfloor = 48$.
32	**(a)** 168 cm^2	Other side $= 24$ cm; area $= 7 \times 24$.
33	**(c)** 14	x-intercept 5, y-intercept 9.
34	**(d)** 2425.5 cm^3	$\frac{2}{3}\pi r^3$.
35	**(a)** 3	Solve $(x - a_0)^2 + b^2 = c^2$.
36	**(b)** 64	Find each term.
37	**(b)** 5	Find the order of 10 modulo q.
38	**(a)** 16	Add 2 to both sides.
39	**(b)** 168	Area $= \frac{1}{2} \times$ chord $\times$ distance.
40	**(b)** 484	Side $= \frac{\text{perimeter}}{4}$.

41	**(a)** Both A and R are true and R is the correct explanation of A.	$48 = 6 \times 8$ and $18 = 6 \times 3$, and no larger number divides both; R is the definition of gcd.
42	**(a)** Both A and R are true and R is the correct explanation of A.	$2 \times 180^\circ = 360^\circ$; R is the standard proof.
43	**(a)** Both A and R are true and R is the correct explanation of A.	$3(2) + 2(3) = 12$, so the pair satisfies the equation; R is the definition used.
44	**(a)** Both A and R are true and R is the correct explanation of A.	$5^3 = 125$, using R.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	A point with abscissa 0 is on the y-axis, so A is true; R is also true but is a statement about the x-axis, so it does not explain A.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Putting $y = 0$ gives $x = 3$, so A is true; putting $x = 0$ gives $y = -6$, so R is true. R gives the other intercept, so it does not explain A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but R is a property of irrational numbers and does not prove that $\sqrt{2}$ is irrational.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	A follows from $a^2 - b^2 = (a - b)(a + b)$; R is true but is a different identity and does not explain A.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true theorems, but R does not by itself explain why equal chords are equidistant.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	$2\pi r = 2 \times \frac{22}{7} \times 7 = 44$, so A is true; R is true but is the area formula, not the reason for A.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	Multiply the fraction by 11 and reduce.
2	**(c)** $(2x+3)(4x^2-6x+9)$	Use the factorisation identity.
3	**(a)** both diameters	Reason with circle theorems.
4	**(d)** four times	Reason with the area formulas.
5	**(b)** 0.80	Reason about probabilities.
6	**(b)** 440	Add the first and last term, multiply by $\frac{n}{2}$.
7	**(c)** 12	Compare numbers of the form $p^a q^b$.
8	**(d)** 60	$\dfrac{72\times 90+144\times 45}{216}=60.$
9	**(b)** 32	Pairs: $(2624,1824)\to(1824,800)\to(1024,800)\to(800,224)\to(576,224)\to(352,224)\to(224,128)\to(128,96)\to(96,32)\to(64,32)\to(32,32)\to(32,0).$
10	**(b)** 131 cm	Each side of the new figure is half a diagonal; there are two of each.
11	**(c)** $-\frac{1}{36}$	$\dfrac{23-(22)}{-24-(12)}=-\frac{1}{36}.$
12	**(c)** 146300 cm^3	$\pi r^2 h$ with $r=35$, $h=38$.
13	**(d)** 72	Area $=\frac{1}{2}\times a\times b$.
14	**(c)** 18	Intercepts are c and $-c/m$; area $=\frac{1}{2}\lvert c\rvert\cdot\lvert c/m\rvert$.
15	**(c)** $\frac{4}{3}$	Each is $\frac{d}{9}$.
16	**(d)** 133	$x^3+y^3=(x+y)^3-3xy(x+y)$.
17	**(c)** 35	Add the two distances from the centre.
18	**(d)** 126	Use Brahmagupta's formula.
19	**(c)** $\frac{1}{9}$	List all pairs with the given product.
20	**(a)** -4	$d=-4$, $a=60$; $t_{17}=a+16d$.
21	**(b)** 34	Squares have an odd number of factors; count the integers whose squares lie in the range.
22	**(c)** 51	$51(100-102)=n_2(98-100)$.
23	**(d)** 15	Trace the sequence step by step.
24	**(c)** 51°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
25	**(b)** 132	Intercepts 11 and 24; area $=\frac{1}{2}\times 11\times 24$.
26	**(b)** 12 cm	$\sqrt{l^2+b^2+h^2}=\sqrt{16+64+64}$.
27	**(b)** 13	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
28	**(d)** 5	Equate the two expressions.
29	**(b)** $\frac{8}{9}$	Add the numerators over 99.
30	**(b)** 117	$x^3-y^3=(x-y)^3+3xy(x-y)$.
31	**(b)** 5	Subtract the two distances from the centre.
32	**(d)** 168	Side of square $=2r$.
33	**(a)** $\frac{5}{12}$	Count the favourable pairs out of 36.
34	**(c)** 70	$t_k=S_k-S_{k-1}=p(2k-1)+q$.
35	**(d)** 17	$n=17$ gives 289, which is composite.
36	**(d)** $6:13$	Amounts are in the ratio $(72-60):(60-34)=12:26$.

37	**(d)** 14	Pairs: $(1971, 1251) \to (1251, 720) \to (720, 531) \to (531, 189) \to (342, 189) \to (189, 153) \to (153, 36) \to (117, 36) \to (81, 36) \to (45, 36) \to (36, 9) \to (27, 9) \to (18, 9) \to (9, 9) \to (9, 0)$.
38	**(b)** 28 cm	Sides 6 cm and 8 cm.
39	**(c)** 1550	Rate $= \dfrac{300}{6} = 50$ per km; fixed charge 100.
40	**(c)** 24948 cm^3	Volume of hemisphere + volume of cone.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R is a general range, so it does not explain why an impossible event gets exactly 0.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	The ratio of consecutive terms of the first is 2 and the difference of consecutive terms of the second is 2; both are true, but they are unrelated statements.
43	**(a)** Both A and R are true and R is the correct explanation of A.	A single counterexample, the number 2, disproves the statement.
44	**(d)** A is false but R is true.	A is false when the groups have different sizes (the weights differ); R is true.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	$29 = 4 \times 7 + 1$, so A is true; $7 = 0 \times 29 + 7$, so R is true; the two statements are unrelated.
46	**(c)** A is true but R is false.	A is a property of parallelograms; R is false (the diagonals of a non-rectangular parallelogram are unequal).
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R describes the shape of the graph but does not by itself say why there are infinitely many solutions.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	TSA $= 6a^2 = 96$, so A is true; R is true but the TSA is $6a^2$, so R does not explain A.
49	**(c)** A is true but R is false.	$x > 0,\ y < 0$ gives Quadrant IV, so A is true. In Quadrant IV the abscissa is positive, so R is false.
50	**(c)** A is true but R is false.	The slope is the coefficient of x, i.e. -4, so A is true; R is false because b is the y-intercept.

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